

1. Business Background

A bank has a portfolio of loan applications covering different borrower and loan characteristics such as state, employment length, loan purpose, home ownership, loan term, grade, verification status, income, DTI, interest rate, and loan status.

The business needs a simple analytical view of this portfolio to understand:

- how much lending activity is present,
- how much money has been funded,
- how the portfolio is performing,
- where lending activity is concentrated,
- and where different segments show different observed loan-performance patterns.

The analysis is designed as a **descriptive portfolio analysis** using the available loan-level data.

2. Business Problem

The bank needs to analyze its loan portfolio to understand **portfolio size, lending activity, loan performance, borrower/loan segment composition, and observed risk patterns**.

The analysis should help the business answer:

Where is the bank lending, how is the portfolio performing, which borrower and loan segments have the most activity or exposure, and which observed patterns should be monitored further?

3. Project Objective

The objective is to use Python to transform the raw loan data into clear and understandable business insights.

The project will:

1. Inspect and understand the available loan data.
2. Identify and handle data-quality issues.
3. Create useful analytical fields.
4. Calculate important portfolio KPIs.
5. Analyze lending activity across major business dimensions.
6. Compare observed Good Loan and Bad Loan performance.
7. Analyze additional risk and pricing patterns.
8. Create clear visualizations.

4. Key Business Questions

Portfolio Performance

- How large is the loan portfolio?
- How much has been funded?
- What loan-level payment amount is recorded?
- What are the average interest rate and DTI?
- What proportion of loans are classified as Good Loans and Bad Loans?

Origination Activity

- How has loan application activity changed over time?
- How has funded amount changed over time?
- Which period has the highest observed activity?

Portfolio Concentration

- Which states have the highest loan activity?
- Which loan terms are most common?
- Which employment-length groups have the highest activity?
- Which loan purposes have the highest activity and funded exposure?
- Which home-ownership categories have the highest activity and funded exposure?

Risk and Loan Performance

- How does observed Bad Loan performance differ across states?
 - How does observed Bad Loan performance differ by loan term?
 - How does observed Bad Loan performance differ by purpose?
 - How does observed Bad Loan performance differ by employment length?
 - How does observed Bad Loan performance differ by home ownership?
 - How does observed performance differ across Grade and Sub Grade?
 - How does observed performance differ by verification status?
 - How does borrower DTI differ across loan-status groups?
 - How does average installment vary by term and loan status?
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5. Good / Bad Loan Definition

For this project, the Good/Bad classification is an analytical rule created because the BRD asks for Good and Bad Loan KPIs but does not explicitly define the mapping.

Loan Status	Analytical Category
Fully Paid	Good Loan
Current	Good Loan
Charged Off	Bad Loan

This classification is used consistently throughout the analysis.

6. Main Analysis Areas

The project is divided into two major analytical parts.

Core BRD & Business Analysis

- Executive Portfolio KPIs
- MTD KPI Analysis
- Good vs Bad Loan KPIs
- Monthly Trends
- State Analysis
- Loan Term Analysis
- Employment Length Analysis
- Loan Purpose Analysis
- Home Ownership Analysis

Additional Risk Insights

- Grade vs Bad Loan %
- Grade vs Interest Rate
- Sub-Grade Risk Drill-Down
- Verification Status vs Bad Loan %
- DTI vs Loan Status
- Term × Loan Status — Average Installment

Each analysis follows:

Business Question → Analysis → Result → Visualisation → Business Interpretation

7. Expected Business Outputs

The final analysis should provide:

- Portfolio-level KPIs
- MTD KPIs
- Good vs Bad Loan metrics
- Monthly lending trends

- State-level activity and observed risk
 - Term-level risk and pricing patterns
 - Employment-length analysis
 - Purpose-level activity and observed risk
 - Home-ownership analysis
 - Grade and Sub-Grade risk/pricing views
 - Verification-status comparison
 - DTI and installment analysis
 - Clear visualizations
 - Key business insights
 - Business recommendations for monitoring and further investigation
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8. Important Data Considerations

The analysis uses only the information available in the supplied loan dataset.

Two important interpretation points are carried throughout the project:

1. **MTD payment limitation:** `total_payment` is a cumulative loan-level payment field, not a transaction-level monthly cash-collection table. Therefore, the MTD payment figure is treated as a loan-level payment proxy for loans originated in the latest observed month.
2. **Partial latest month:** the latest Issue Date is **12-Dec-2021**, so December MTD figures represent the observed period through 12-Dec-2021 rather than a complete month.

Observed differences between segments are descriptive and should be interpreted as **observed portfolio patterns**, not as proof of causation.

9. Final Business Goal

The goal of this project is not simply to calculate numbers.

The goal is to move from:

Raw Loan Data

→ **Clean & Structured Data**

→ **Business KPIs**

→ **Segment Analysis**

→ **Risk & Pricing Insights**

→ **Visualisation**

→ **Business Interpretation**

→ **Monitoring & Investigation Actions**

This provides the business with a simple, reproducible view of the loan portfolio and gives the analyst a clear framework for explaining the findings to stakeholders.

10. Project Mind Map

